Module 1 knowledge check

Your AI app stores conversation metadata with varying structures. Which PostgreSQL data type best supports this?

A. TEXT
B. JSONB
C. VARCHAR(MAX)

Your app creates many short-lived database connections during AI inference. What approach improves performance?

A. Open a new connection for each message and close immediately
B. Use a ConnectionPool to maintain reusable connections
C. Keep a single global connection open for the entire app lifetime

Module 2 knowledge check

Which pgvector distance operator should you use when your embeddings are normalized to unit length and you want to measure semantic similarity?

A. <=> (cosine distance)
B. <-> (L2 distance)
C. <#> (negative inner product)

You're building a RAG pipeline that needs to retrieve relevant document chunks quickly from 5 million embeddings. The collection receives occasional batch updates but no real-time inserts. Which index type should you choose?

A. IVFFlat with an appropriate number of lists
B. HNSW with high ef_construction value
C. No index, relying on exact sequential scan

Module 3 knowledge check

You need a vector index for 5 million product embeddings with daily full refreshes. Build time must be under 30 minutes. Which index should you choose?

A. IVFFlat with lists set to sqrt(rows)

B. HNSW with m=16 and ef_construction=64

C. HNSW with m=8 and ef_construction=32

Your filtered vector search query filters by category_id then orders by similarity. The query plan shows a sequential scan. What should you check first?

A. Verify a B-tree index exists on category_id

B. Verify the vector index uses the same operator class as the query

C. Increase hnsw.ef_search to expand the search space